A. The Best Card

time limit per test 2 seconds
memory limit per test 512 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy, math, number theory | https://codeforces.com/problemset/problem/2253/A

In a card game, there are n cards with values 2, 3, 4, ..., $n + 1$.

To determine which of two cards with values x and y wins, apply the following rules:

- if one of the numbers x and y is divisible by the other, the card with the smaller value wins;
- otherwise, the card with the larger value wins.

For example, between cards 2 and 6, card 2 wins because 6 is divisible by 2. Between cards 4 and 6, card 6 wins because neither of these numbers is divisible by the other.

Determine whether there exists a card that wins against every other card.

Input

The first line contains an integer t ($1 \le t \le 10^4$) — the number of test cases.

The only line of each test case contains an integer n ($2 \le n \le 2 \cdot 10^5$) — the number of cards in the game.

Additional constraints on the input:

- the sum of n over all test cases does not exceed $3 \cdot 10^6$.

Output

For each test case, print YES if there is a card that wins against all other cards, and NO otherwise.

Each letter may be printed in either case. For example, YES, yes, and yEs are all recognized as a positive answer.

Example

Input
5 2 3 4 5 8

Output
YES NO YES NO NO

Note

In the first test case, the available cards have values 2 and 3. Card 3 wins against card 2.

In the second test case, the available cards have values 2, 3, and 4. Card 2 wins against card 4, card 3 wins against card 2, and card 4 wins against card 3, so there is no suitable card.